

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Working with Vectors

Vector questions become simple after choosing components and keeping direction signs visible.

WORKING WITH VECTORS · M1 · 02

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The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 COMP: Component Form

TRIGGER *bearing, magnitude, vector components*

02 RESULT: Resultant And Bearing

TRIGGER *resultant force, magnitude, bearing*

03 PATH: Position Path

TRIGGER *position vector, velocity vector*

04 CLOSE: Closest Distance

TRIGGER *minimum distance, nearest approach*

BANK EVIDENCE Local pages: 23 problem, 46 answer, 162 notes. Website primary entries: 27. Website linked entries: 42.

01 COMP: Component Form

TRIGGER *bearing, magnitude, vector components*

BECOMES Resolve the vector into i and j components.

FIRST LINE TO WRITE

$$\mathbf{v} = (v_E)\mathbf{i} + (v_N)\mathbf{j}$$

SIMPLEST STRATEGY

1. Choose east/north axes.
2. Obtain the reference angle.
3. Multiply by sine/cosine.
4. Place signs by quadrant.

WORKED MODEL

▮ A speed 40 on bearing 150° is $20\mathbf{i} - 20\sqrt{3}\mathbf{j}$.

HARD VARIANTS

1. East component uses sine of the bearing; north uses cosine.
2. Check signs from the quadrant, not from memory.
3. For angles from the i -axis, the sine/cosine roles change.

— BOTTOM LINE

Bearings are measured clockwise from north.

COMMON TRAP Swapping sine and cosine for bearings.

02 RESULT: Resultant And Bearing

TRIGGER *resultant force, magnitude, bearing*

BECOMES Add components, then find magnitude and direction.

FIRST LINE TO WRITE

$$\mathbf{R} = \sum \mathbf{F}, \quad |\mathbf{R}| = \sqrt{x^2 + y^2}$$

SIMPLEST STRATEGY

1. Resolve every vector.
2. Enter signs.
3. Sum components.
4. Use Pythagoras.
5. Locate quadrant.
6. Turn angle into bearing.

WORKED MODEL

| $\mathbf{R} = -4\sqrt{3}\mathbf{i} - 14\mathbf{j}$ points south-west.

HARD VARIANTS

1. Bearings must be written as three digits.
2. If the vector is west/south, add the reference angle to 180° or 360° correctly.
3. Use exact components until the final angle.

— BOTTOM LINE

A bearing answer must match the quadrant.

COMMON TRAP Giving the acute angle but not the bearing.

03 PATH: Position Path

TRIGGER *position vector, velocity vector*

BECOMES Build position as start plus velocity times time.

FIRST LINE TO WRITE

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$$

SIMPLEST STRATEGY

1. Place the initial vector.
2. Add velocity times time.
3. Tidy i and j components.
4. Handle units of time.

WORKED MODEL

■ $\mathbf{r} = (60 - 20t)\mathbf{i}$ for motion west from $60\mathbf{i}$.

HARD VARIANTS

1. Use the time unit matching the velocity unit.
2. Meeting means both components are equal at the same time.
3. Due north/south/east/west means one component is fixed.

— BOTTOM LINE

Position vector is a moving address.

04 CLOSE: Closest Distance

TRIGGER *minimum distance, nearest approach*

BECOMES Minimise distance squared.

FIRST LINE TO WRITE

$$d^2 = x(t)^2 + y(t)^2$$

SIMPLEST STRATEGY

1. Create separation vector.
2. Length squared.
3. Open the quadratic.
4. Square complete or use vertex.
5. Evaluate distance.

WORKED MODEL

$$(t - 4)^2 + (5 - t)^2 = 2(t - 4.5)^2 + 0.5.$$

HARD VARIANTS

1. Keep $d^2 \geq 0$; the minimum distance is not the minimum squared value unless square-rooted.
2. Reject a minimum time outside the allowed time interval.
3. For two particles, use one separation vector, not two separate distances.

— BOTTOM LINE

Minimise d^2 , then square-root at the end.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
bearing, magnitude, vector components	$\mathbf{v} = (v_E)\mathbf{i} + (v_N)\mathbf{j}$
resultant force, magnitude, bearing	$\mathbf{R} = \sum \mathbf{F}, \quad \mathbf{R} = \sqrt{x^2 + y^2}$
position vector, velocity vector	$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$
minimum distance, nearest approach	$d^2 = x(t)^2 + y(t)^2$

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
COMP: Component Form	East component uses sine of the bearing; north uses cosine. Check signs from the quadrant, not from memory. For angles from the i -axis, the sine/cosine roles change.
RESULT: Resultant And Bearing	Bearings must be written as three digits. If the vector is west/south, add the reference angle to 180° or 360° correctly. Use exact components until the final angle.
PATH: Position Path	Use the time unit matching the velocity unit. Meeting means both components are equal at the same time. Due north/south/east/west means one component is fixed.
CLOSE: Closest Distance	Keep $d^2 \geq 0$; the minimum distance is not the minimum squared value unless square-rooted. Reject a minimum time outside the allowed time interval. For two particles, use one separation vector, not two separate distances.